

Assignments

This course includes four progressive assignments that will guide you through building a **Fitness Tracker App** — a real-world mobile application for tracking physical activities. Each assignment builds upon the previous one, gradually incorporating more advanced technologies and features using **React Native** and **Expo**.

Overview

The Fitness Tracker App is designed for users to log and manage their physical activities, such as running or swimming, while maintaining data across app sessions. The app allows users to:

- Log and manage activity metadata (type, duration, calories)
- Navigate across multiple screens using **React Navigation** or **Expo Router**
- Manage global state using **Context API** or **Redux Toolkit**
- Persist data locally using **React Native Async Storage**

Across four assignments, you will gain hands-on experience with modern mobile development technologies and best practices in the React Native ecosystem, progressing from component-based design to full state management and data persistence.

Assignment Timeline

Each assignment builds upon the previous one, so start early and maintain a steady progress throughout the course.

Assignment	Topic	Release Date	Due Date
1	Basic App Structure	September 14	September 28
2	Styling and Navigation	September 28	October 12
3	Navigation and State Management	October 12	October 26
4	Advanced State Management and Persistence	October 26	November 9

Note

- All assignments are due at **11:59 PM EST** on the due date
- Each assignment contributes **12.5%** to your final grade
- Assignments must be submitted through Quercus
- Start early to allow time for questions and debugging

Assignment Structure

Assignment 1: Basic App Structure

Build the foundational structure of the Fitness Tracker App using React Native and Expo.

Key Features:

- Log activities with metadata (type, duration, calories)
- Display activities in a list using `FlatList`
- Update activities from the list
- Delete activities from the list
- Input validation and state management with `useState`

Technologies:

- React Native
- Expo
- TypeScript
- Core components (`View` , `Text` , `TextInput` , `Button` , `FlatList`)

Assignment 2: Styling and Navigation

Enhance the app with professional styling and multi-screen navigation.

Key Features:

- Style the app using `StyleSheet` and Flexbox
- Centralize styles and colors for consistency
- Implement type-safe navigation with React Navigation (Home, Add/Edit Activity, Activity Details screens)
- Pass data between screens

Technologies:

- React Native
- Expo
- TypeScript
- React Navigation
- `StyleSheet` and Flexbox

Assignment 3: Navigation and State Management

Enhance the Fitness Tracker App with Expo Router for navigation and Context API for global state management.

Key Features:

- Implement navigation with Expo Router
- Use dynamic routes for activity details (e.g., `[id].tsx`)
- Manage activities using Context API (add, update, delete)
- Validate user inputs and display error messages

Technologies:

- React Native
- Expo
- TypeScript
- Expo Router
- Context API

Assignment 4: Advanced State Management and Persistence

Extend the Fitness Tracker App by integrating Redux Toolkit for centralized state management and React Native Async Storage for data persistence.

Key Features:

- Set up a global store using Redux Toolkit (`@reduxjs/toolkit`)
- Create slices for managing activities (add, update, delete)
- Connect Redux store to the app with `Provider`
- Persist activities using React Native Async Storage
- Automatically load saved data on app startup
- Validate user inputs and display error messages

Technologies:

- React Native
- Expo
- TypeScript
- Expo Router
- Redux Toolkit
- React Native Async Storage

Submission Guidelines

- All assignments must be submitted through Quercus.
- Each assignment has specific submission requirements detailed in its handout.
- Auto-grading will be used to evaluate submissions.

Questions and Support

- Check assignment specifications carefully before starting.
- Use [GitHub Discussions](#) ↗ for assignment-related questions.
- Attend office hours for additional support.

Last updated on September 1, 2025